Systems Analysis & Design

Semester 2026-I

Workshop No. 3

Systems Analysis - Local business support recovery

Group Seven

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1. Executive Summary

This project presents the design and enhancement of a support system for small businesses in the Titan Plaza shopping mall, based on the results obtained in Workshops 1 and 2. The initial analysis identified key issues such as low conversion of visitors into sales, limited visibility of local businesses, and the lack of tools for data-driven decision-making. In response, a platform was proposed to integrate customer, sales, and inventory data in order to improve business management and overall performance.

In Workshop 3, the system is refined through the application of robust design principles, ensuring scalability, maintainability, and reliability. Additionally, risk management strategies, quality assurance practices, and project planning are incorporated to guarantee a feasible and effective implementation in a real-world environment.

2. Robust Architecture Design

2.1 Architectural Overview

The system is designed using a layered architecture that separates responsibilities into different levels to improve organization, scalability, and sustainability. It is composed of three main layers: the presentation layer (frontend), the application layer (backend), and the data layer (database), along with an integration layer for external services.

The frontend provides interfaces for business owners and customers, enabling interaction with the system. The backend handles the core logic, including data processing, analysis, and system operations. The database stores all persistent information, such as user data, sales records, and inventory data.

2.2 Design Principles and Quality Standards

The system architecture incorporates key software engineering principles to ensure robustness and long-term sustainability. Modularity is applied by dividing the system into independent components, allowing for easier updates and maintenance. Scalability is considered to support future business expansion or even deployment in other shopping malls without requiring major redesigns.

Maintainability is ensured through the clear separation of layers and well-defined responsibilities within the system. Fault tolerance is addressed through the implementation of offline functionality and data synchronization mechanisms to handle connectivity issues within the shopping mall.

Additionally, the design follows quality standards based on recognized practices such as ISO 9001.

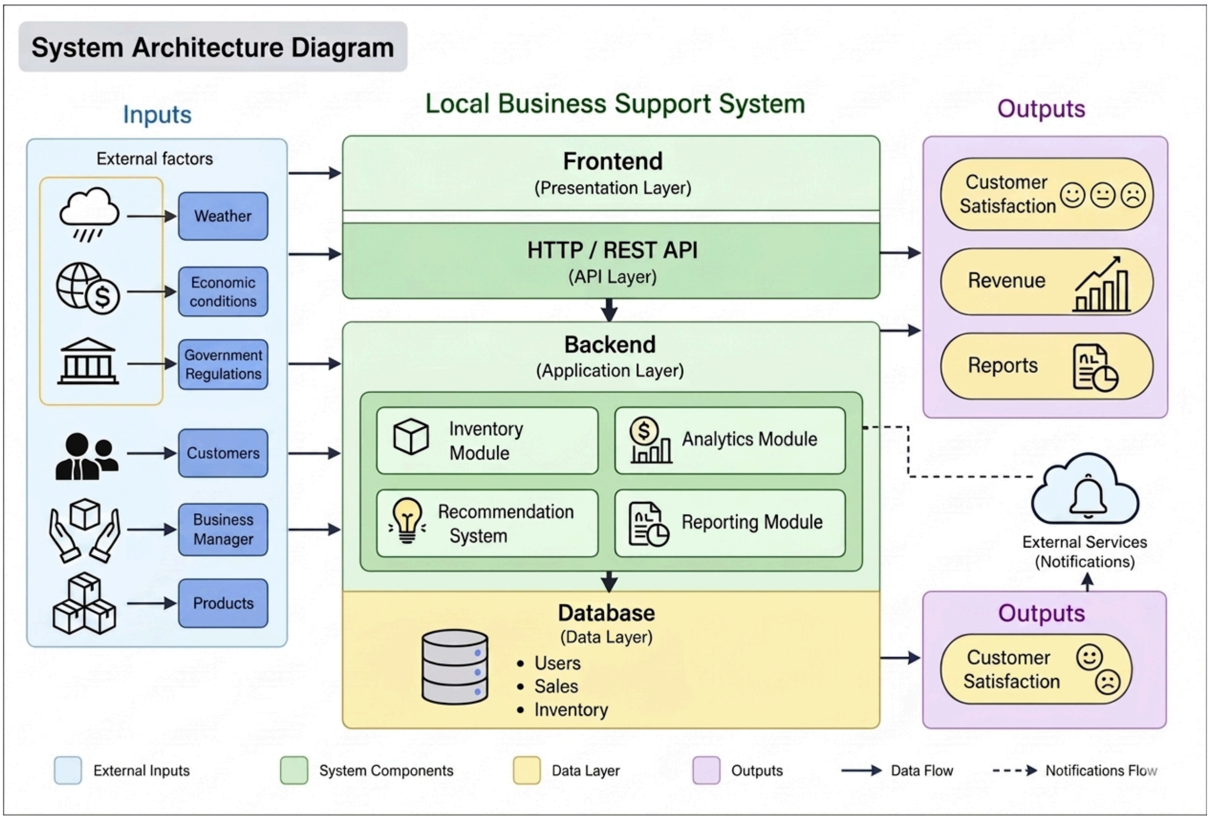
2.3 System Components and Integrations

The system is composed of several key components that interact to support business operations. These include the user interface, business management module, inventory management module, analytics module, and recommendation system. Each component fulfills a specific function while maintaining communication with others through the backend.

Integration between components is achieved through well-defined interfaces and API-based communication. The frontend communicates with the backend through HTTP requests, while the backend interacts with the database and internal modules to process and store information. External services are also integrated to enhance system capabilities, such as sending notifications and enabling data synchronization.

This modular and integrated structure ensures efficient data flow, flexibility, and the ability to adapt to future requirements.

2.4 System Architecture Diagram



3. Risk Management Plan

3.1 Risk Identification and Analysis

During the system design, several risks were identified that may affect its proper operation and implementation. These risks include technical, operational, and user-related factors. Among the main risks are the dependence on external factors such as weather and day of the week, as well as the decrease in consumer spending.

Additionally, risks related to connectivity within the shopping mall, the potential resistance of business owners to adopt the system and share information, and the possibility of data inconsistencies were identified. Each of these risks can directly impact the effectiveness of the system and is therefore considered a key aspect of the analysis.

3.2 Risk Mitigation Strategies

To reduce the impact of the identified risks, several mitigation strategies are proposed. First, data analysis mechanisms are implemented to anticipate customer behavior patterns and improve decision-making for businesses.

Regarding connectivity issues, the system includes offline functionalities and synchronization processes that allow it to continue operating without relying on a stable connection. To address user resistance, a training process and an intuitive interface are proposed to facilitate system adoption.

Additionally, contingency strategies are considered, such as data validation to prevent inconsistencies and the use of notifications to alert users about potential inventory or sales issues, ensuring a clear response to any adverse situation.

3.3 Monitoring and Control Mechanisms

The system incorporates monitoring mechanisms that allow continuous evaluation of its performance and the effectiveness of the implemented strategies. Periodic tracking of system usage by business owners will be conducted, along with key indicators such as sales growth, promotion effectiveness, and prediction accuracy.

Furthermore, connectivity and system errors will be logged to identify potential failures and enable continuous improvements. These control mechanisms allow the system to be progressively adjusted, ensuring its proper functioning and adaptation to real-world conditions.

3.4 Risk Matrix

Risk	Probability	Impact	Level	Mitigation

				Strategy
Dependence on external factors (weather, day of week)	High	High	Critical	The owner selects the day and weather manually (or uses a free API). The dashboard then shows a red warning: "PREPARE INVENTORY"
Decrease in consumer spending	High	High	Critical	The system adds loyalty points ($10,000 = 1 point). The owner sees a table of customers with the most points and can offer them a discount
Owner resistance to adopt the system	Medium	High	High	Invite 2 store owners to try the system for free. If they like it, they will recommend it to others
Data inconsistencies	Medium	Medium	Medium	If the owner types a product that is not in the list, the system shows: "Product not found. Do you want to create it?"

4. Quality Assurance Framework

4.1 Quality Objectives and Metrics

The system defines a set of quality objectives aimed at ensuring proper functionality, usability, and performance. These objectives include system availability, interface response speed, and the accuracy of the generated analyses and recommendations.

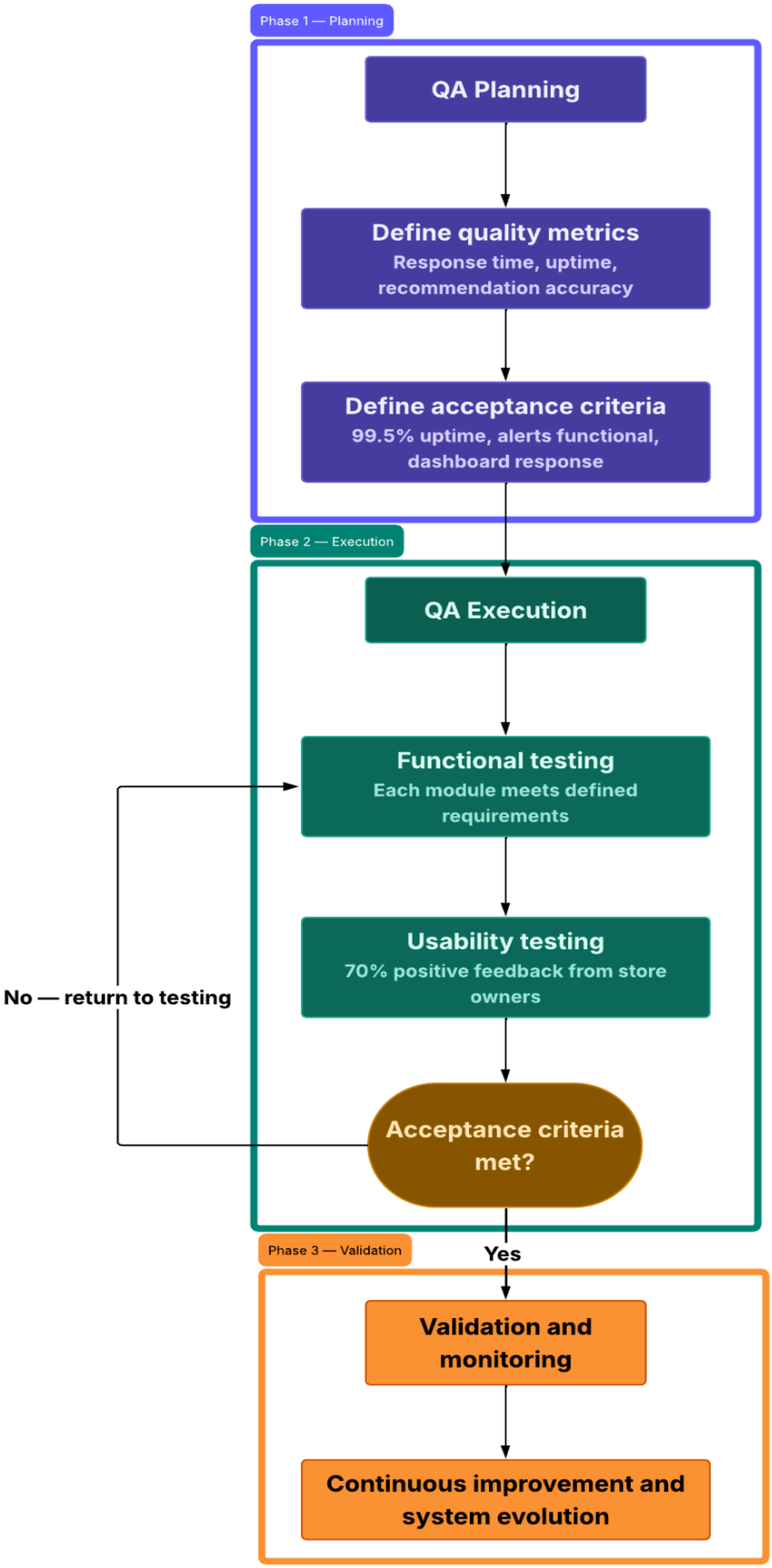
To evaluate these objectives, metrics such as system response time, availability rate during operating hours, and the effectiveness of the predictions are established. These metrics allow the measurement of system performance and ensure that it meets the defined requirements.

4.2 Validation and Acceptance Criteria

The system will be validated through different testing strategies that verify its correct functionality before implementation. These include functional tests to ensure that each module fulfills its purpose, as well as usability tests to guarantee that users can interact with the system in a simple and intuitive way.

Acceptance criteria are defined based on the fulfillment of system requirements, such as adequate response times, correct data visualization in the dashboard, and reliable operation of alerts and recommendations. Meeting these criteria ensures that the system is ready for implementation in a real-world environment.

4.3 Quality Assurance Process Flow


Phase 1 — Planning
QA Planning
Define quality metrics
Response time, uptime, recommendation accuracy
Define acceptance criteria
99.5% uptime, alerts functional, dashboard response
Phase 2 — Execution
QA Execution
Functional testing
Each module meets defined requirements
Usability testing
70% positive feedback from store owners
Acceptance criteria met?
No — return to testing
Yes
Phase 3 — Validation
Validation and monitoring
Continuous improvement and system evolution

5. Implementation Strategy

5.1 Implementation Phases and Deployment

The system implementation will be carried out using a phased approach, allowing for progressive and controlled development. In the first phase, the core structure of the system will be established, including the database and the main business logic. The second phase will focus on the development of functional modules such as sales recording, inventory management, and the alert system.

In the third phase, analytical components will be incorporated, including the recommendation system and prediction models based on customer behavior. Finally, in the fourth phase, a pilot deployment will be conducted with a small group of businesses within the shopping mall, followed by a full rollout once the results have been validated.

5.2 Infrastructure and Requirements

The implementation of the system requires a technological infrastructure that supports the operation of its different layers. This includes servers for the backend, a database for data storage, and a web interface accessible from both mobile and desktop devices.

Additionally, the system relies on integration with external services through APIs for functionalities such as sending notifications and managing real-time data. These integrations enhance system capabilities and improve interaction with users.

6. Project Management

6.1 Project Charter

The Local Business Recovery Support System addresses a structural informational disconnect identified at Titan Plaza shopping mall, where small independent businesses lack the data infrastructure necessary to compete effectively against large retail chains. This project develops an analytical platform that centralizes sales, inventory, and customer behavior data to enable data-driven decision-making for small business operators.

Project Scope

In scope: design, development, and phased deployment of the analytics platform for small businesses within Titan Plaza, including the owner dashboard, inventory alert system, demand forecasting module, recommendation engine, and automated promotions feature.

Out of scope: large chain anchor stores, e-commerce integrations beyond API connectivity, and mall administrative systems.

Project Objectives

- Achieve a minimum of 70% adoption rate among small business owners during the pilot phase.
- Reduce inventory stockout incidents by at least 30% within the first three months of full deployment.
- Maintain 99.5% system uptime during mall operating hours.

Stakeholders:

Stakeholder	Role	Interest
Small business owners	Primary users	Improved sales and decision-making
Store employees	Secondary users	Simplified inventory and sales recording
Mall administrators	Operational context	Mall attractiveness and tenant success
Customers	Indirect beneficiaries	Better product availability and promotions
Development team	Builders	Successful delivery of the system
Course instructor	Academic supervisor	Engineering rigor and documentation quality

Success Criteria

- Platform delivered on schedule across all four implementation phases.
- Owner dashboard operational with real-time metrics during pilot deployment.
- Inventory alert system validated against actual stockout events.
- System compliant with Colombian personal data protection regulations (Law 1581 of 2012).
- Positive usability feedback from at least 70% of pilot participants.

Constraints and Assumptions

- Development must operate within academic resource constraints.
- Internet connectivity in some mall zones requires offline synchronization capability.
- Some store owners may be resistant to adopting new technology, requiring dedicated onboarding support.
- Integration with existing third-party inventory systems must not disrupt current workflows.

6.2 Team Structure and Roles

The project team is composed of four members, each assigned to specific responsibilities that reflect their participation across the three workshops. Given the academic nature of the project, roles are distributed to ensure balanced contribution while maintaining clear accountability for each area of the project.

Organizational Structure

The team operates under a flat collaborative structure, where all members share decision-making responsibility. Each member leads a primary domain while supporting the others across overlapping tasks.

Member	Primary Role
Daniel Eduardo Rincón Vivas	Systems Analyst
Sara Sofia Socha Gil	Project Coordinator
Diego Alejandro Yañes Zabala	Design & Documentation Lead
Sebastian Zambrano Hurtado	Risk & Quality Lead

Role Descriptions

Systems Analyst — Responsible for ensuring the proposed system solution remains aligned with the findings and requirements identified during the analysis phase. Leads the review and synthesis of system requirements, oversees the coherence between analytical findings and design decisions, and contributes to the validation of system components against identified user needs.

Project Coordinator — Responsible for maintaining the overall project structure, tracking progress across workshops, and ensuring deliverables are completed on time and to the required standard. Coordinates team communication, manages the project timeline, and serves as the primary point of contact for academic and stakeholder reporting.

Design & Documentation Lead — Responsible for the clarity, organization, and professional quality of all project documentation and visual representations. Leads the preparation of diagrams, reports, and presentation materials, ensuring that technical content is communicated effectively to both technical and non-technical audiences.

Risk & Quality Lead — Responsible for identifying potential risks across the project lifecycle and proposing mitigation strategies. Leads quality assurance activities, monitors compliance with established standards, and ensures that acceptance criteria and validation approaches are properly defined and tracked throughout the project.

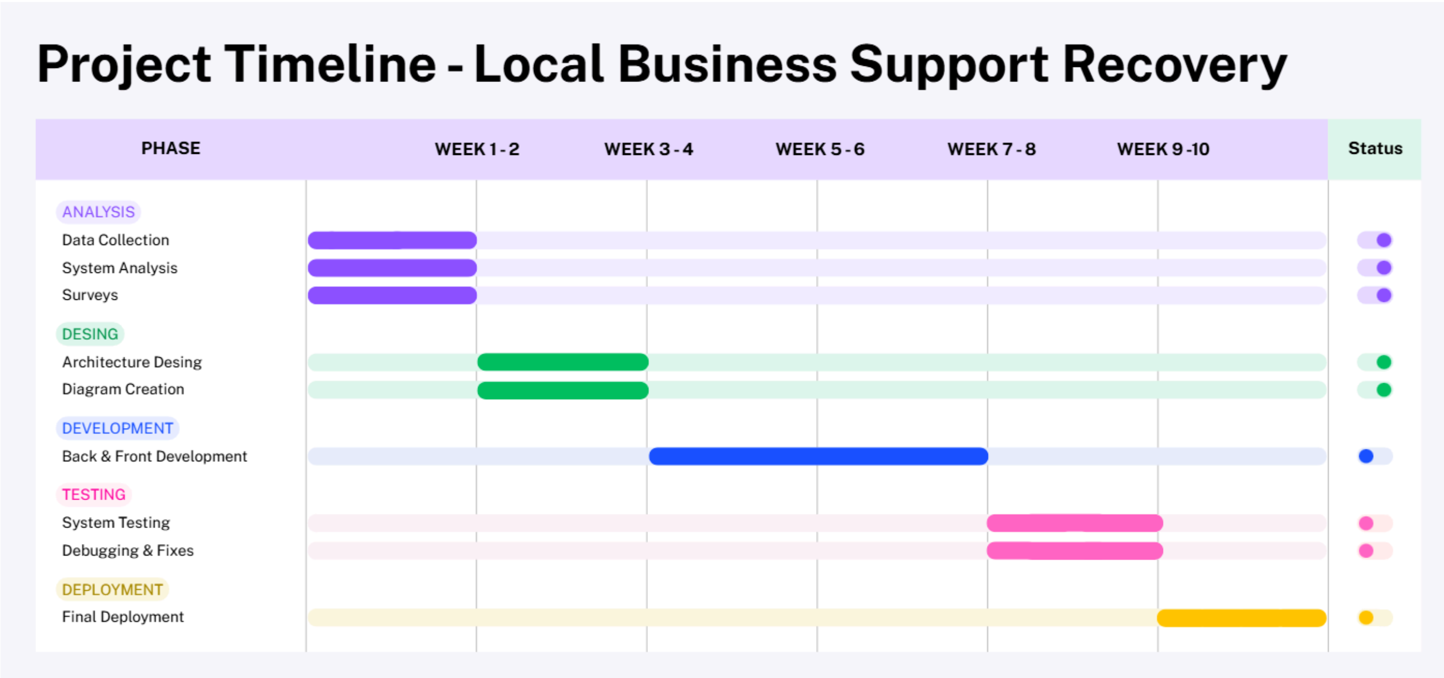
Shared Responsibilities

All team members share responsibility for the following activities regardless of their primary role:

- Participation in data collection and field observation sessions
- Review and validation of all workshop deliverables before submission
- Contribution to risk identification and mitigation discussions
- Attendance and preparation for academic presentations

Maintenance of the GitHub repository and project documentation

6.3 Project Timeline



Milestones	Status
WORKSHOP 1	Completed
WORKSHOP 2	Completed
WORKSHOP 3	In Progress
WORKSHOP 4	Upcoming
CATCH-UP 1	Upcoming
FINAL DELIVERY	Upcoming

6.4 Resource Management

The project is developed by a team of four students whose contributions are distributed according to the roles defined in section 6.2. Each member dedicates time across all project phases, with workload intensity varying depending on the activities corresponding to their primary role. Effort is expected to increase during phase transitions and deliverable submission periods, and the team coordinates workload distribution on a weekly basis to prevent bottlenecks and ensure balanced participation throughout the project lifecycle.

Technological Resources

The project relies on openly available and academic tools to minimize cost while maintaining professional quality standards.

Resource	Purpose
GitHub	Version control and repository management
Lucidchart	Architecture and flow diagrams
Google Workspace	Documentation and spreadsheets
Overleaf	Technical report preparation in LaTeX
Canva	Slides and presentations

Budget Considerations

Given the academic nature of the project, all development tools and platforms used are either open source or available under free academic licenses. The primary resource investment is the time contributed by team members. No external financial budget is required for the current academic phase of the project. Should the platform advance toward a real deployment beyond the academic context, budget considerations would need to address cloud infrastructure costs, potential licensing fees for enterprise tools, and operational support expenses.

6.5 Communication and Control Plan

Communication Plan:

Weekly meetings will be held to review progress and define tasks. Daily communication will be conducted via messaging platforms to maintain constant coordination. Additionaly github will be used to share changes documentation and maintain version control.

Control Plan:
Project progress will be monitored through milestones and periodic reviews. GitHub will be used for change tracking. System quality will be ensured through testing, validation, and bug fixes. Problems will be identified and resolved collaboratively and evaluations will be conducted at each phase to ensure objectives are met

Monitoring and Improvement:
The project will be continuously monitored to identify areas for improvement and implement adjustments as needed ensuring the system meets the established objectives.

7. Evolution Summary

7.1 Design Evolution

The development of this project has followed a progressive process, from the initial analysis to the definition of a system ready for implementation. In Workshop 1, an analysis of the real environment was conducted, identifying the main issues affecting small businesses in the Titan Plaza shopping mall, such as external factors influencing sales, variability in customer flow, and the lack of technological tools.

In Workshop 2, these findings were transformed into a proposed solution through the design of a structured system architecture, defining its components, modules, and data flow. This established a solid foundation for system development, focused on improving decision-making and optimizing business management.

Finally, in Workshop 3, the design was refined through the incorporation of software engineering principles, risk management strategies, quality assurance, and project planning. This allowed the solution to evolve into a robust, viable system prepared for implementation in a real-world environment.

7.2 Lessons Learned and Improvements

Throughout the development of the project, important lessons were learned regarding the importance of understanding the real context before designing a technological solution. It became evident that external factors, such as customer behavior and environmental conditions, significantly influence business performance.

Additionally, the need to design flexible and scalable systems capable of adapting to different conditions and evolving over time was identified. The incorporation of monitoring, validation, and continuous improvement mechanisms ensures that the system not only addresses current needs but can also adapt to future changes.

These improvements reflect an evolution in the understanding of the problem and in the ability to propose more comprehensive solutions aligned with real-world conditions.

8. References

1. ISO. (2015). ISO 9001:2015 Quality Management Systems — Requirements. International Organization for Standardization.

2. IEEE. (2014). Guide to Software Engineering Body of Knowledge (SWEBOK). IEEE Computer Society.
3. Pressman, R. S., & Maxim, B. R. (2020). Software Engineering: A Practitioner's Approach (9th ed.). McGraw-Hill.

4. Sommerville, I. (2016). Software Engineering (10th ed.). Pearson.

5. Fowler, M. (2002). Patterns of Enterprise Application Architecture. Addison-Wesley.

6. Microsoft. (2023). REST API Guidelines. Retrieved from https://learn.microsoft.com/en-us/azure/architecture/best-practices/api-design

7. Firebase. (2023). Firebase Cloud Messaging Documentation. Retrieved from https://firebase.google.com/docs/cloud-messaging